Anatomy of the heart and mediastinum

2nd year

Block CVS

Prepared by

Prof. Dr. Faten Youssif Mahmoud
Faculty of Medicine –Assiut University

2022-2023

Pericardium

The pericardium is a sac that encloses the heart and the roots of the great vessels. It lies within the middle mediastinum (Fig.1). It consists of a an outer fibrous layer called (**the fibrous pericardium**) and an inner serous sac called (**the serous pericardium**) (Fig. 2).

The fibrous pericardium: It is a fibrous sac encloses the heart and also the roots of the great blood vessels passing through it. Its broad **base** overlies the central tendon of the diaphragm and firmly attached to it. **Anteriorly,** it is attached to the sternum by the sternopericardial ligaments. **On each side**, the pericardium is related to the phrenic nerves, pleura and lung (Fig.1). At the sternal ends of the fourth and fifth intercostal spaces, the pericardium is directly related to the thoracic wall where the left pleura and lung deviat laterally forming the cardiac notch. This part is called the bare area of the pericardium**. Posteriorly**, the pericardium is related to the descending aorta and oesophagus which separate it from the middle four thoracic vertebrae.
It is supplied with the pericardiophrenic and internal thoracic arteries.

The serous pericardium: It is a closed serous sac inside the fibrous pericardium which invaginated by the heart. It is formed of **parietal** and **visceral layers**. **The parietal layer** lines the fibrous pericardium and is reflected around the roots of the great vessels to become continuous with the visceral layer of serous pericardium. **The visceral layer** is closely applied to the heart and it is called the epicardium. The space between the parietal and visceral layers is called

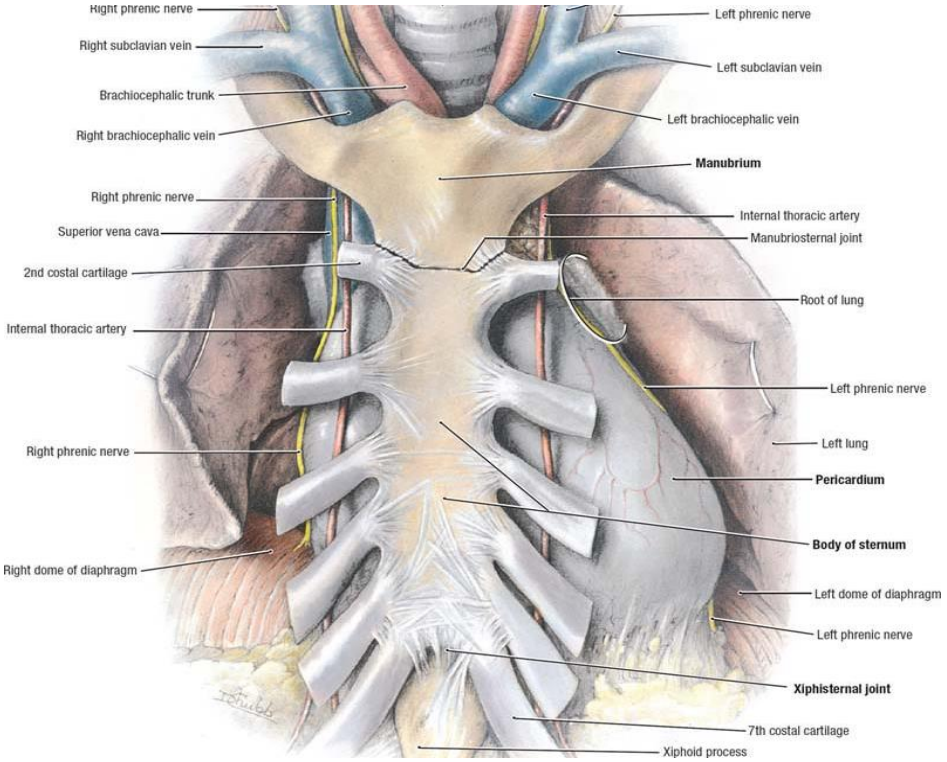


Fig.(1): It shows the position of the pericardium and heart in the thoracic cavity

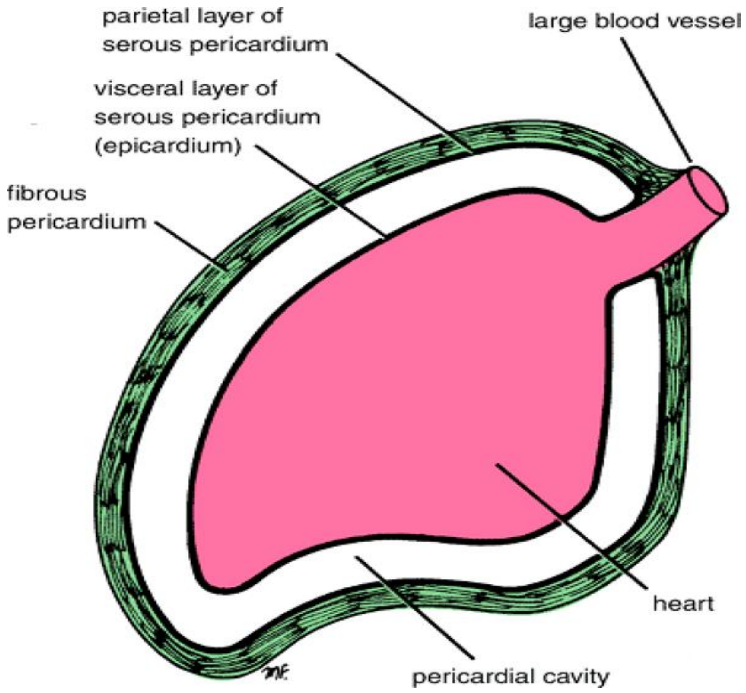


Fig.(2): It shows the different layers of the pericardium

the pericardial cavity. The cavity contains a small amount of fluid which acts as a bubricant to facilitate movements of the heart .

Pericardial sinuses: There are two sinuses between the parietal and visceral layers of the pericardium, the transverse sinus and the oblique sinus of the pericardium. They are formed as a result of irregularity of the line of reflection of the parietal pericardium (Fig. 3).

Transverse sinus of the pericardium: It lies between the ascending aorta and the pulmonary trunk (anteriorly) and the right and left atrium (posteriorly). In order to demonstrate it, pass your finger transversely between the ascending aorta and the superior vena cava on the right side until it appears on the left side between the pulmonary trunk and the left auricle. It is separated from the oblique sinus by the reflection of the pericardium between the right and left superior pulmonary veins.

Oblique sinus of the pericardium: It represents a cul-de-sac in the pericardial cavity posterior to the heart. It is formed by an arch of reflected parietal pericardium. It lies between the left atrium (anteriorly) and the fibrous pericardium lined by parietal layer of serous pericardium (posteriorly). The sinus is bounded on the right side by the inferior vena cava and right pulmonary veins. On the left side, it is bounded by the left pulmonary veins. Superiorly, the sinus is closed by the reflection of visceral layer from the back of left atrium onto the fibrous pericardium. Inferiorly, the orifice of the sinus opens into the rest of the pericardial cavity.

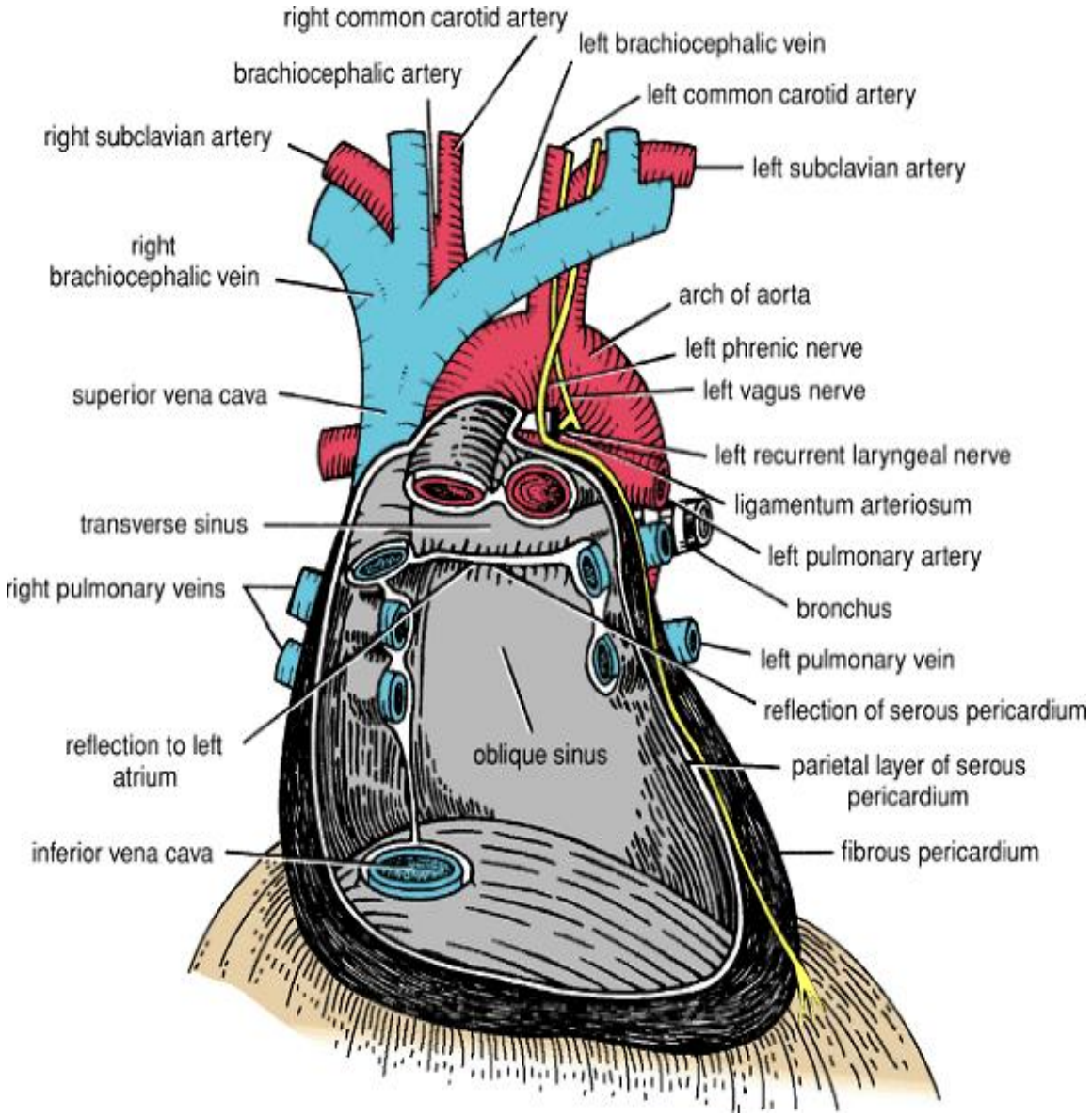


Fig. (3): It demonstrates the sinuses of the pericardium.

Heart

The heart is a hollow muscular organ pyramidal in shape. It lies inside the pericardium occupying the major part of the middle mediastinum. It is formed of four chambers, the right and left atria and the right and left ventricles.

External features of the heart:

The heart has an anterior (sternocostal) surface, an inferior (diaphragmatic) surface and a posterior surface (base). It also has an apex. The surfaces of the heart are separated by four borders (upper, right, left and lower border).

The sternocostal surface: This surface is related to the sternum and the costal cartilages from the third to the sixth on both sides. It is also related to the pleura and lung on both sides. The sternocostal surface is directed forwards, upwards and to the left. This surface is formed of two portions, an aterial portion and a ventricular portion. **The atrial portion** is formed mainly by the right atrium and by the auricle of the left atrium. The **ventricular portion** is formed mainly by the right ventricle which represent about 2/3 of this surface and the left ventricle which represent about 1/3 of the surface. The line of separation between the two ventricles is marked by the anterior interventricular groove which lodges the anterior interventricular artery. The part of the atrioventricular groove on the sternocostal surface lodges the right coronary artery (Fig. 4).

The diaphragmatic surface: This surface is related to the central tendon and left cupola of the diaphragm. The heart rests on its diaphragmatic (inferior) surface. This surface is directed downwards and

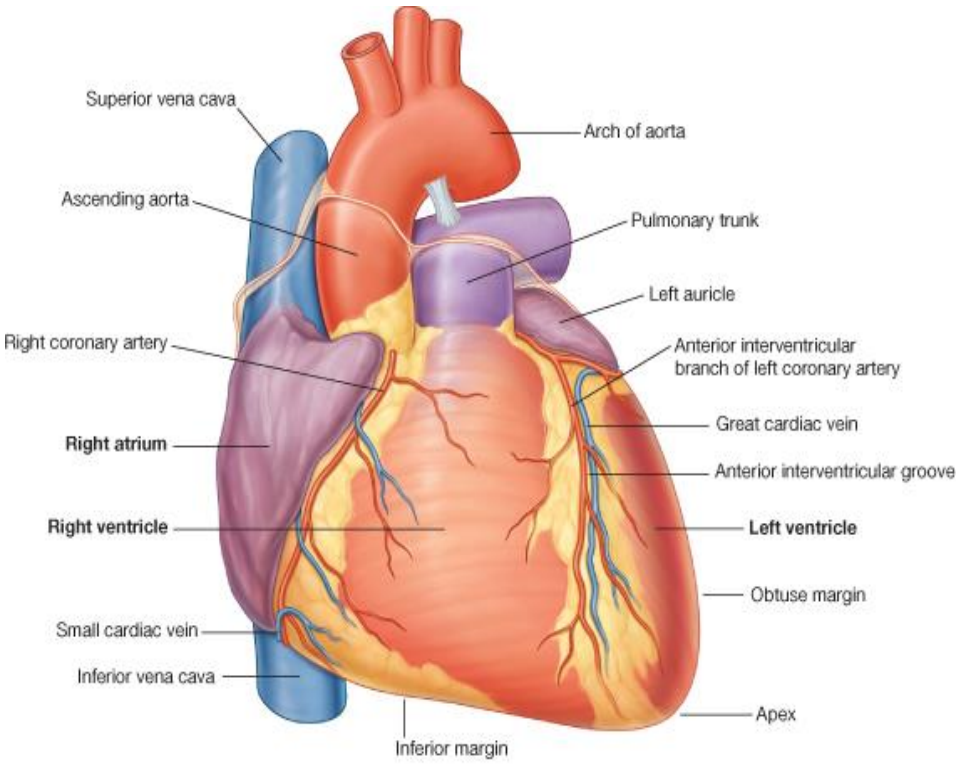


Fig. (4): Anterior (sternocostal) surface of the heart.

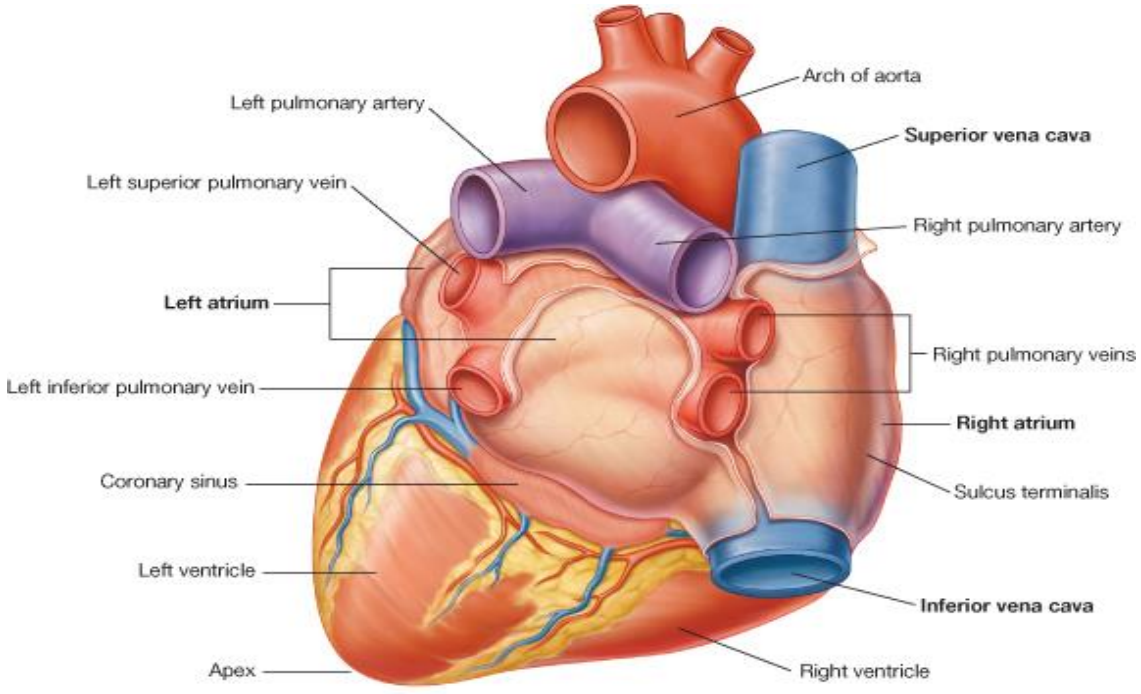


Fig. (5): Inferior (diaphragmatic) surface of the heart.

backwards. It is formed mainly by the left ventricle which represents about 2 / 3 of the surface and only one third by the right ventricle. The two ventricles are separated by the posterior interventricular groove which lodges the posterior interventricular artery and the middle cardiac vein. This surface is separated from the base of the heart by the posterior part of the atrioventricular groove which lodges the coronary sinus (Fig.5).

The base of the heart (the posterior surface): It is directed backwards and to the right. It is formed mainly by the left atrium and to a small extent by the back of the right atrium. It lies opposite to the thoracic vertebra from the sixth to the ninth in the erect position (Fig. 6).

The apex of the heart: It is formed by the left ventricle .It is directed downward, forward and to the left. It lies at level of the left fifth intercostal space 3 ½ 2 inches (9 cm) from the median plane.

Note: In the region of the apex, the apex beat can usually be seen and palpated in the living subject.

Borders of the heart:

The upper border: It is formed by the two atria, mainly the left. It is hidden anteriorly by the pulmonary trunk, ascending aorta and superior vena cava.

The right border: It is formed by the right atrium

The lower border: It is formed by the right ventricle.

The left border: It is formed by the left ventricle and the left auricle.

Note:

The limits between the chambers of the heart are marked on the surface by grooves or sulci. The atria are separated from the ventricles

by the atrioventricular or coronary groove (sulcus) which lodges the main trunks of the coronary arteries. The two ventricles are separated by the anterior and posterior interventricular grooves.

Surface marking of the heart

In order to outline the heart, it is considered to have an apex and four borders (Fig.7).

Apex of the heart: It is located in the left fifth intercostal space 3 ½ inches (9 cm) from the median plane.

Note: The apex of the heart can be felt below and medial to the nipple of the left breast.

The upper border: It extends from a point on the second left costal cartilage 1 ½ inch from the median plane to a point at the third right costal cartilage 1 inch from the median plane.

The right border: It extends from a point on the third right costal cartilage 1 inch from the median plane (the previous point) downward to a point on the 6th right costal cartilage ½ inch from the median plane.

The lower border: It extends from a point on the 6th right costal cartilage ½ inch from the median plane (the previous point) to a point 3 ½ inches from the median plane in the left fifth intercostal space (the apex of the heart).

The left border: It extends from the point of the apex of the heart to a point on the second left costal cartilage 1 ½ inches from the median plane.

Surface marking of the atrioventricular groove (coronary groove) : It is represented by a line extends between the 3rd left sternocostal junction to the 6th right sternocostal junction.

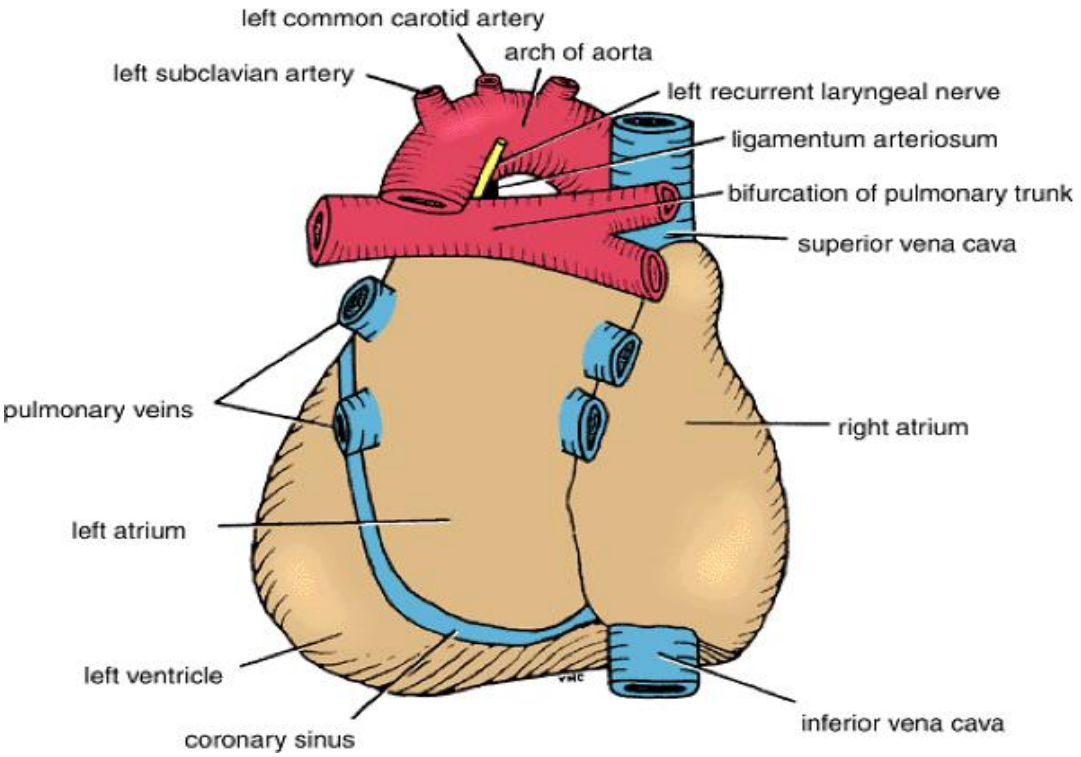


Fig.(6): Posterior surface(base) of the heart.

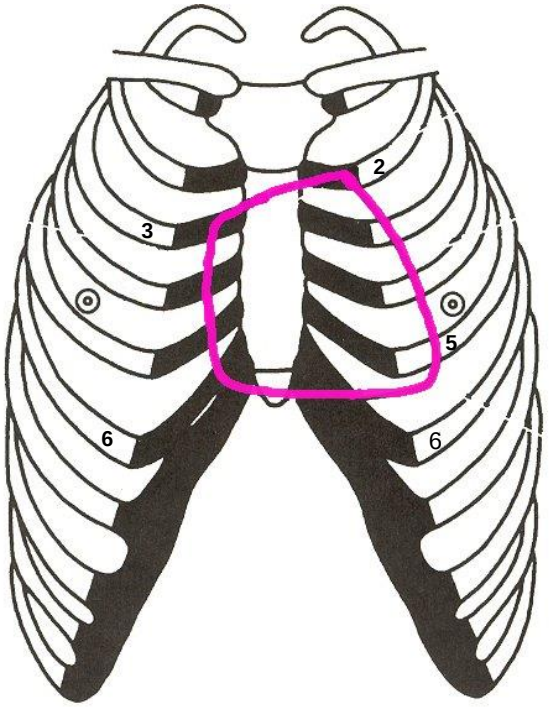


Fig. (7): Surface marking of the heart.

Cardiac chambers

Right atrium

The right atrium has the thinnest wall of the four chambers. The crista terminalis divides this chamber into two parts (a posterior part and an anterior part).

I) **The posterior part** has a smooth wall and receives the openings of the great veins.

II) The anterior part has rough wall due to the presence of a muscular ridges called the **musculi pectinate.** These ridges extend forward from the crista terminals on the anterior wall of the right atrium. They run parallel to each other. This part of the right atrium is called the **atrium proper.** Thia part is continuous anteriorly with the **right auricle** which has a muscular wall.

Note:

The crista terminalis is a smooth longitudinal muscular ridge on the lateral wall of the right atrium. It extends from the right margin of the orifice of the superior vena cava to the right margin of the orifice of the inferior vena cava. The crista terminalis gives rise to the musculi pectinati that run across the wall of the atrium proper.

The atrial septal wall separates right and left atria. The interatrial septum presents an oval depression called the **fossa ovalis** which lies above and to the left of the opening of the inferior vena cava. This fossa is bounded by a margin called the **annulus ovalis** or limbus fossae ovalis. This margin is prominent above and on the sides and it is ill- defined below (Fig.8).

Note:

The fossa ovalis is the embryologic remnant of the fetal foramen ovale.

Openings into the right atrium.

- **The superior vena cava** opens into the upper posterior part of the right atrium. It has no valves.
- **The inferior vena cava** opens into the lower posterior part of the right atrium. It is guarded by a rudimentary nonfunctioning valve.
- **The coronary sinus** opens into the right atrium between the inferior vena cava and the tricuspid valve. It is guarded by a valve that covers the lower part of the opening and prevents regurgiation of blood into the coronary sinus during contraction of the right atrium.

Openings of the anterior cardiac veins:

They are three or four small veins which collect blood from the front of the right ventricle and open into the anterior part of the right atrium.

Venae cordis minimi: They are numerous minute veins that opens directly into the atria and to a less extent into the ventricle.

Note:

Blood leaves the right atrium into the right ventricle through the right atrioventricular (tricuspid) orifice.

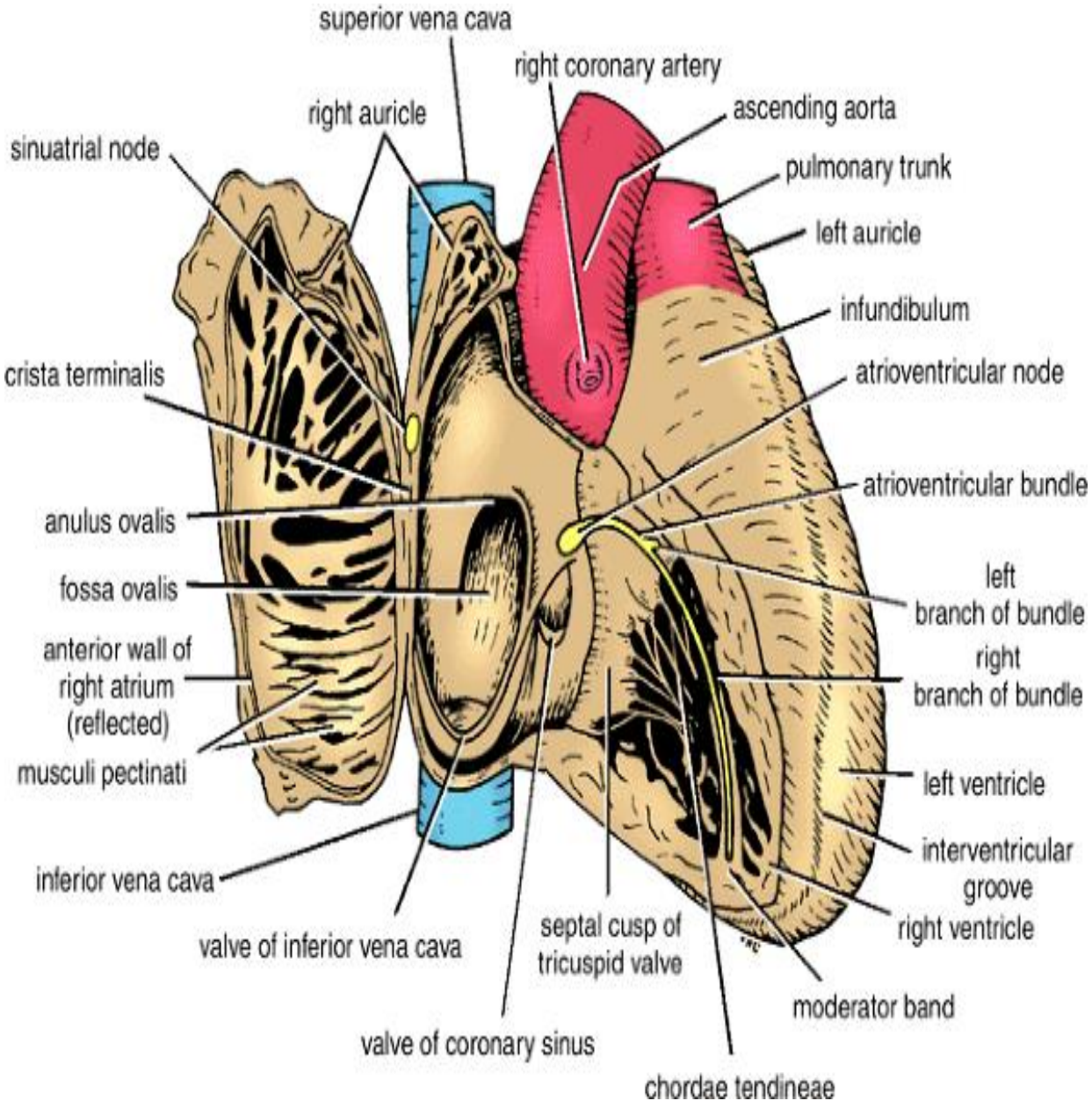


Fig.(8): Chambers of the right side of the heart .

The anterior heart wall has been dissected away revealing the right atrium and the right ventricle.

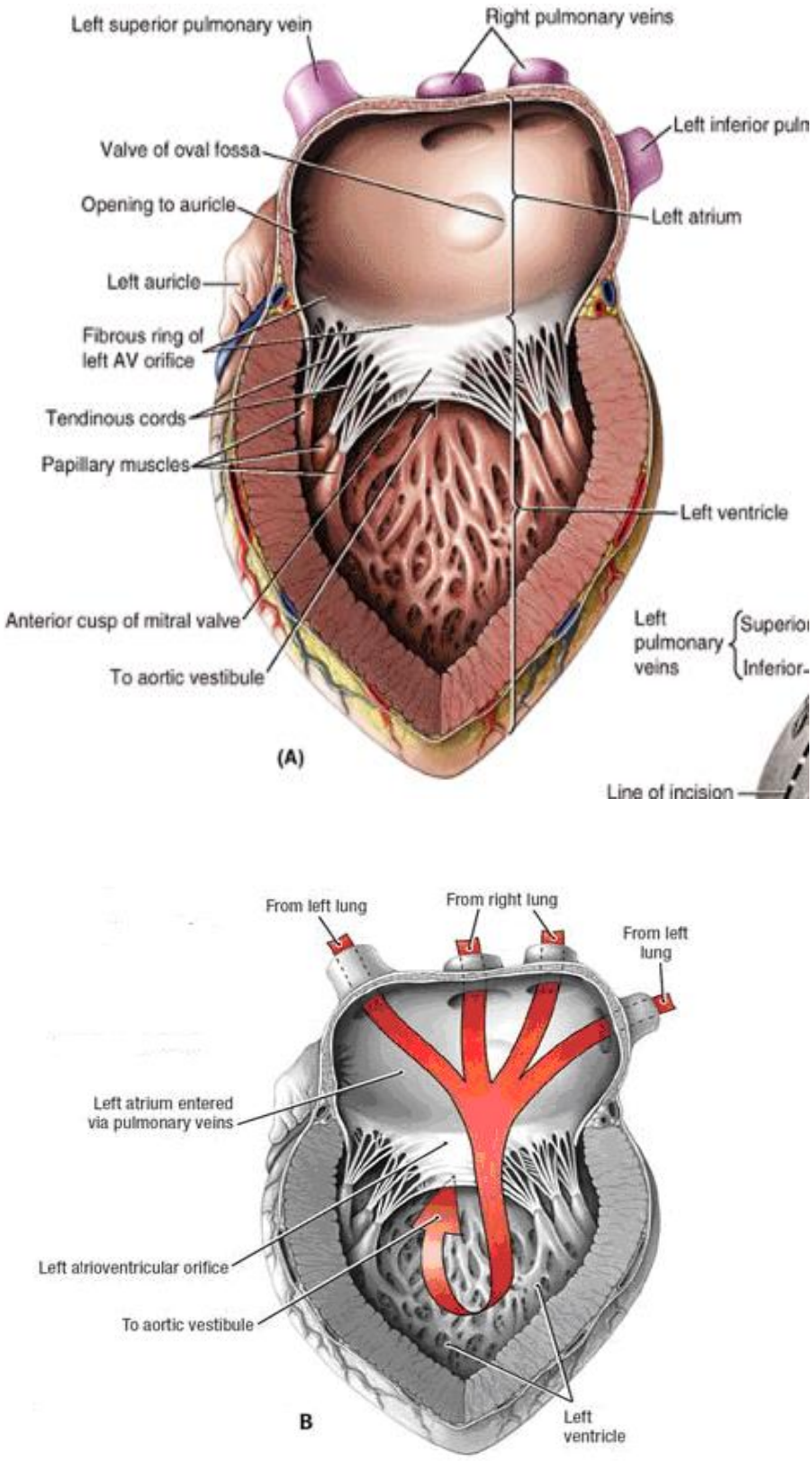


Fig. (9A&B): Chambers of the left side of the heart .

The heart wall has been dissected away revealing the left atrium and the left ventricle.

Left atrium:

This chamber has slightly thicker walls than the right atrium because increased effort is required during atrial systole to overcome the elasticity of the extremely thick left ventricular wall. The left atrium consists of a main cavity and an auricle.

- The interior of the left atrium is smooth but its auricle has few and small musculi pectinati (Fig.9).
- The interatrial septum may present an impression corresponding to the fossa ovalis of the right atrium.

Note:

The left atrium is situated behind the right atrium and forms the greater part of the base of the heart. Behind it lies the oblique sinus of the serous pericardium and the fibrous pericardium separates it from the esophagus.

Openings into the left atrium:

- Four pulmonary veins (two right and two left) open into the upper part of its posterior wall. They have no valves. They carry the oxyganted blood from the lungs to the left atrium.
- Several minute veins (venae cordis minimi) open into the cavity of the left atrium

Blood leaves the left atrium into the left ventricle through the left atrioventricular (mitral) orifice.

Right ventricle:

This chamber has a moderately thick wall to overcome the pulmonary vascular resistance. The **supraventricular crest** is a muscular ridge lies between the tricuspid and pulmonary orifices. It divides the right ventricle into two parts, the **right ventricle**

proper (inflowing part) and **the infundibulum (outflowing part).**

I The right ventricle proper:

It lies below the level of the supraventricuar crest. It is called the inflowing part as it receives non-oxygenated blood from the right atrium through the tricuspid orifice.

The wall of the right ventricle proper is rough due to the presence of muscular projections known as trabeculae carnae. Some of these muscular projections are large and called the papillary muscles. The right ventricle shows the presence of the following papillary muscles (Fig.8).

A) **Anterior papillary muscle** which arises from the anterior wall of the right ventricle .

B) **Posterior papillary muscle** which arises from the posterior wall.

C) **Septal papillary muscles** which they are a number of small muscles arising from the ventricular septum .

Each papillary muscle has a base continuous with the ventricular wall and an apex which gives origin to the chordae tendinae which are connected to the cusps of the tricuspid valve.

Note: *There is a muscular band called the septomarginal trabecula or the moderator band connects the ventricular septum with the base of the anterior papillary muscle across the lumen of the right ventricle near to the apex. The function of this band is to transmit the right branch of the atrioventricular bundle and to prevent over-distension of the right ventricle.*

II The infundibulum of the right ventricle:

It lies above the level of the supraventricular crest. It has a smooth wall. It is called the outflowing part as it leads to the pulmonary orifice.

Left ventricle:

The wall of the left ventricle is three times as thick as the wall of the right ventricle to overcome the vascular resistance of the systemic vascular bed. The left ventricle is divided into two regions (the left ventricle proper and the aortic vestibule).

I) The left ventricle proper: Its interior is rough due to the presence of the trabeculae carnae as in the right ventricle but they are more dense especially at the apex of the heart. It shows the presence of two large and thick papillary muscles (an anterior and a posterior one). Each one sends its chordae tendinae to both cups of the mitral valve (Fig.9).

II) The aortic vestibule: It has a smooth wall and lies superior and to the right of the mitral valve. It leads into the ascending aorta and contains the aortic valve.

The left ventricle receives oxygenated blood from the left atrium through the left atrioventricular (mitral) orifice and leaves it through the aortic orifice.

Note

The interventricular septum separates the two ventricles. It is oblique in position. The anterior border of the septum corresponds to the anterior interventricular groove. Its posterior border corresponds to the posterior interventricular groove. The septum is also curved being concave towards the left ventricle and convex towards the right ventricle. The septum is composed of two parts (a large lower muscular portion and a small upper membranous portion). The membranous part is a common site for the ventricular septal defect.

Valves of the heart

The tricuspid valve (Right atrioventricular valve):

This valve guards the right atrioventricular orifice (tricuspid orifice) which connect the right atrium with the right ventricle. The tricuspid valve has three cusps (anterior, posterior and septal). Each cusp is formed of two endothelial layers with a layer of fibrous tissue in between. The bases of the cusps are attached to the fibrous ring of the skeleton of the heart and to the bases of the adjoining cusps. Their free edges and the ventricular surfaces are attached to the chordae tendineae. The chordae tendineae connect the cusps to the papillary muscles. The papillary muscle is attached by its chordae to two adjoining cusps. The large anterior papillary muscle is connected by chordae to the anterior and posterior cusps, a smaller posterior papillary muscle is attached to the posterior and septal cusps and the septal small papillary muscles join the septal and anterior cusps (Fig.10). So, when the right ventricle contract, the cusps come near to each other and close the orifice.

The mitral valve (left atrioventricular valve) (bicuspid valve):

This valve guards the mitral orifice (left atrioventricular orifice) which connects the left atrium with the left ventricle. It has two cusps (anterior and posterior). The anterior cusp is large and lies anterior and to the right. The posterior cusp is small and lies posterior and to the left. The bases of the cusps are attached to the margins of the atrioventricular orifice. Each cusp receives chordae tendinae from both papillary muscles of the left ventricle

(Fig.10). The mitral valve prevents regurgitation of blood into the left atrium during ventricular systole.

Note: *The right atrioventricular orifice is wider than the left atrioventricular orifice. The right atrioventricular orifice admits the tips of three fingers while the left atrioventricular orifice admits the tips of two fingers only.*

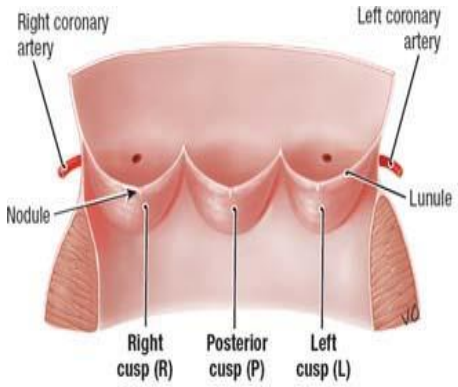
Pulmonary valve:

It guards the pulmonary orifice. The pulmonary orifice is a circular aperture at the summit of the infundibulum leading to the pulmonary trunk. Its diameter is about 3 cm. it lies close to the ventricular septum, above and to the left of the tricuspid valve. The pulmonary orifice lies at a higher level than the aortic orifice.

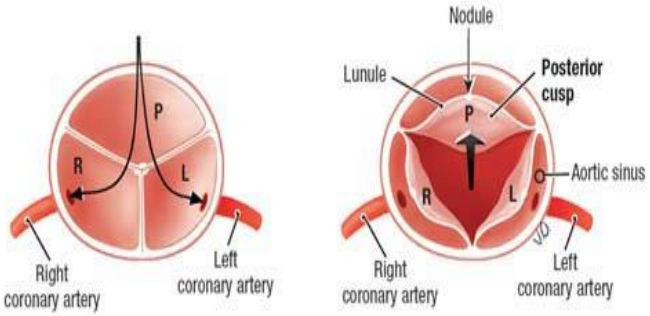
The pulmonary valve consists of three semilunar cusps. Each cusp is formed by folds of endocardium with some fibrous tissue enclosed.The fibrous tissue is thickened in the middle of the free margin to form a nodule .The free margin of the cusp forms two crescentic thin bands (lunules) one on each side of the nodule.

The curved lower margin of each cusp is attached to the arterial wall. The free edges of the cusps are directed upward into the pulmonary trunk. The three semilunar cusps are arranged as one posterior in position and two anterior in position. At the root of the pulmonary trunk are three dilatations called the sinuses. One is situated external to each cusp.

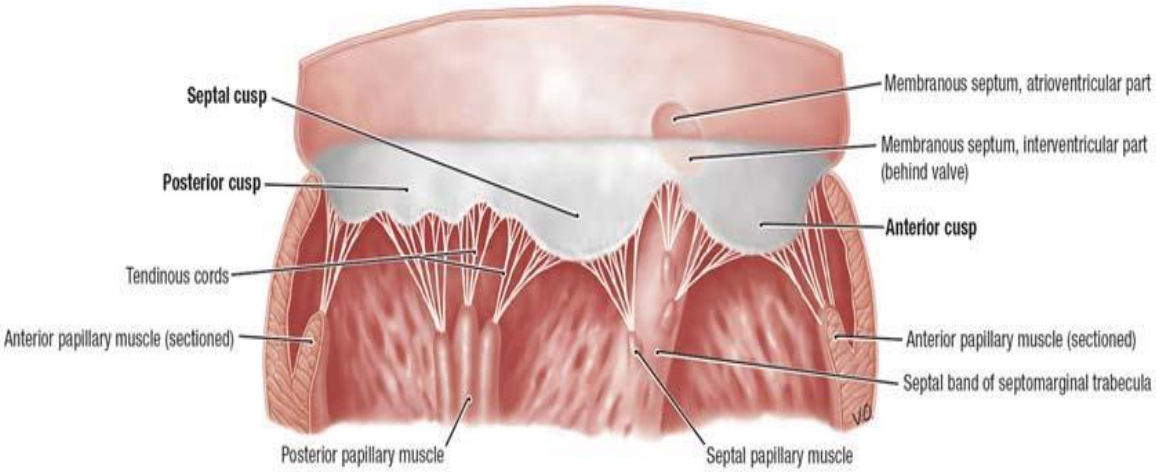
The pulmonary semilunar valve prevents regurgitation of ejected blood (reverse flow from the pulmonary trunk into the ventricle during ventricular diastole).



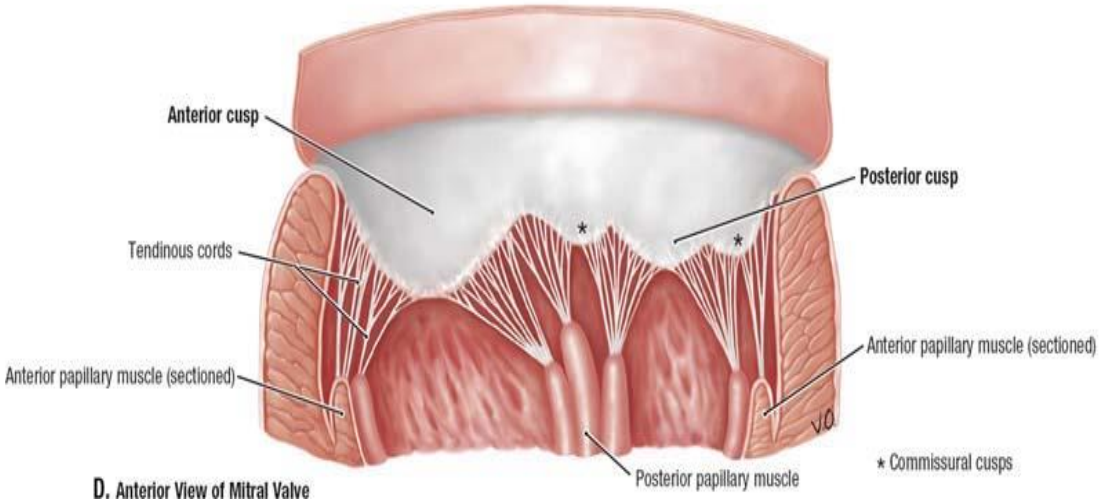
A. Left Posterior Oblique View of Aortic Valve



B. Superior Views of Aortic Valve (Arrows indicate direction of blood flow)



C. Anterior View of Tricuspid Valve



D. Anterior View of Mitral Valve

Fig.(10A, B,C&D) :Valves of the heart .

The aortic valve:

It guards the aortic orifice which it is a circular aperture about one inch in diameter. It present in the aortic vestibule of the left ventricle leading to the ascending aorta (Fig 10)

The aortic valve consists of three semilunar cusps which they are similar in structure to those of the pulmonary valve. However, the arrangement of the cusps is different from that of the pulmonary valve.

One cusp is situated on the anterior wall and two cusps are located on the posterior wall. The root of the ascending aorta presents three dilatations, one opposite each cusp. These are called the aortic sinuses and they are larger than the corresponding pulmonary sinuses. The anterior aortic sinus gives origin to the right coronary artery and the left posterior sinus gives origin to the left coronary artery.

The aortic valve prevents regurgitation of blood from the aorta into the left ventricle during ventricular diastole.

Surface anatomy of the heart valves (Fig.11)

- **The tricuspid valve** lies behind the right half of the sternum opposite the fourth intercostals space.
- **The mitral valve** lies behind the left half of the sternum opposite the fourth costal cartilage.
- **The aortic valve** lies behind the left half of the sternum opposite the third intercostals space.
- **The pulmonary valve** lies behind the medial end of the third left costal cartilage and the adjoining part of the sternum.

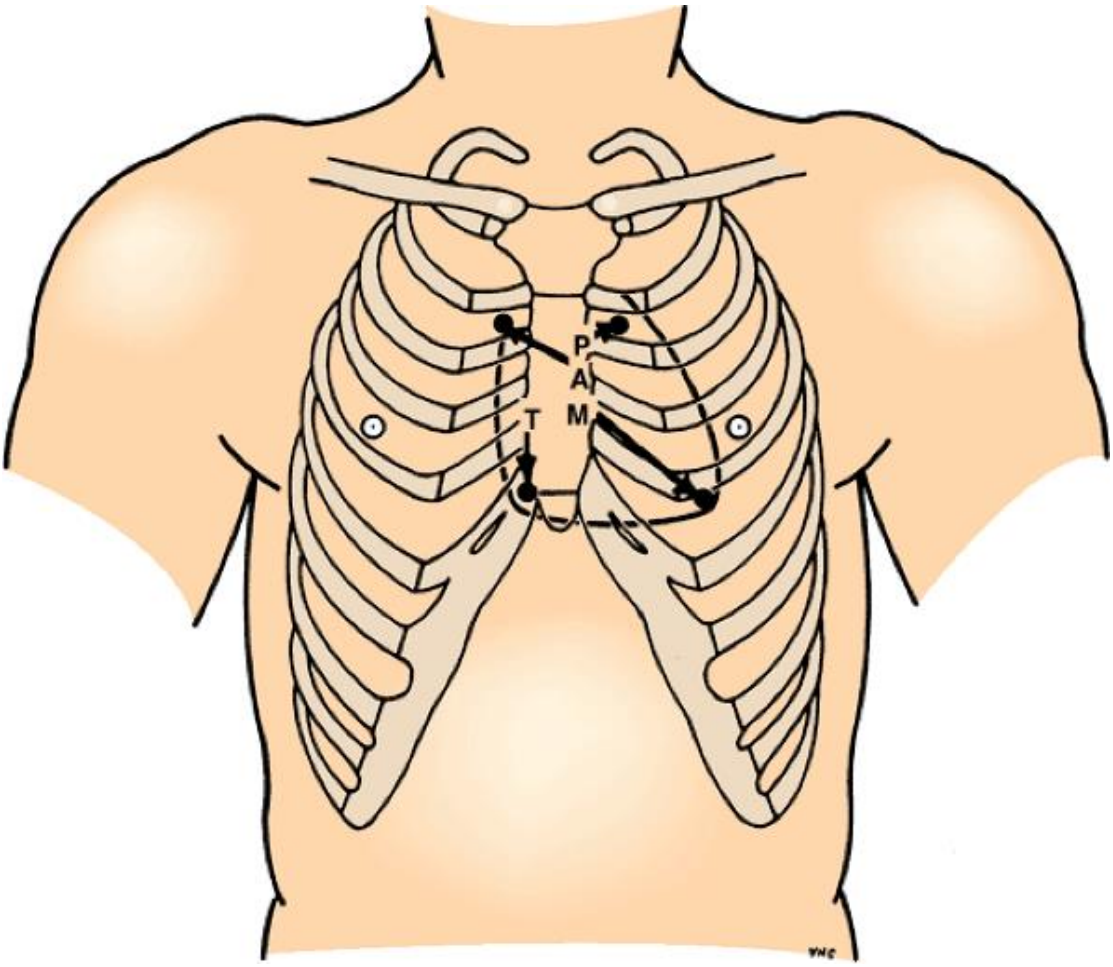


Fig. (11): Surface marking of the heart valves .

(A) aortic valve, (M) mitral valve, (T) tricuspid valve, (P) pulmonary valve. Arrows indicate positions where valves may be heard with least interference

Blood supply of the heart

A) Arterial supply of the heart (Fig. 12):

The heart receives its arterial supply from **the right and left coronary arteries** which arise from the aortic sinuses of the **ascending aorta** just above the cusps of the aortic valve.

The right coronary artery: It arises from the anterior sinus of the ascending aorta. It passes between the right auricle and the root of the pulmonary trunk. Then, it passes downward and to the right in the coronary groove, between the atrial and ventricular portions of the sternocostal surface. It curves on the lower border of the heart and runs in the coronary groove to anastomose with the left coronary artery.

Branches of the right coronary artery:

1- The sinoatrial (S. A. nodal) branch: It is the first major branch in about 60 % of the individuals. It supplies the upper part of the right atrium and the S. A. node.

2- Marginal branch: It runs along the lower border of the heart and supplies the right ventricle.

3- Posterior interventricular artery: It runs in the posterior interventricular groove and supplies both ventricles and the posterior third of the ventricular septum. Near the apex of the heart it anastomoses with the anterior interventricular artery.

4- The atrioventricular (A.V) nodal artery: It arises from the right coronary artery near its termination.

5- The terminal branches, of the right coronary artery usually continue a few centimeters beyond the posterior

interventricular sulcus and anastomose with the termination of the left coronary artery in the coronary groove.

The left coronary artery:

It arises from the left posterior sinus of the ascending aorta. It is larger than the right coronary artery. It passes to the left behind the pulmonary trunk and then between the pulmonary trunk and the left auricle. Then, it curves on the left border of the heart and continues in the coronary groove where it ends by anastomosing with the right coronary artery.

Branches of the left coronary artery:

1- The circumflex branch: It is considered the continuation of the left coronary artery. It curves on the left border of the heart and continues in the coronary groove. It supplies the left atrium and the base of the left ventricle.In about 90 % of the individuals, a large branch called the left marginal artery arises from it supplying the left ventricle . It ends by anastomosing with the right coronary artery.

2- Anterior interventricular artery: It is a large branch that descends in the anterior interventricular groove. It gives branches to both ventricles and the anterior two third of the ventricular septum. At its termination it anastomoses with the posterior interventricular artery near the apex of the heart.

3- In about 40 % of the individuals, it gives an S. A. nodal branch.

Notes:

- *There is a considerable variation in the branches of the coronary arteries and the areas supplied by each vessel among the population.*

- *In healthy young individuals, most of the arteries are **true end-arteries**, which supply discrete volumes of the myocardium. If any anastomoses between left and right coronary arteries are present, they are inadequate o maintain effective circulation in the event of sudden-onset occlusion of a major branch of the coronary circulation. In the slow-onset coronary occlusion as in the atherosclerotic coronary artery disease, anastomoses develop to an extent that collateral circulation is relatively effective.*

B) **Veins of the heart**:

Most of the venous return of the heart passes to the coronary sinus. The **coronary sinus** is about an inch long lies in the posterior part of the coronary groove between the left atrium and the left ventricle (Fig.13). It opens into the right atrium between the opening of the inferior vena cava and the right atrioventricular valve. It receives the following tributaries.

1-The great cardiac vein: It lies in the anterior interventricular groove. It drains the anterior portion of the interventricular septum and the anterior aspects of both ventricles. It also drains the left atrium.

2-The middle cardiac vein: It lies in the posterior interventricular groove. It drains the posterior aspects of both ventricles and the posterior portion of the interventricular septum.

3-The small cardiac veins: It runs along the lower margin of the heart. It drains the right ventricle and the right atrium.

4-Oblique vein of the left atrium: It is a small vein which descends obliquely on the back of the left atrium to join the coronary sinus. It is the embryologic remnant of the left common cardinal vein.

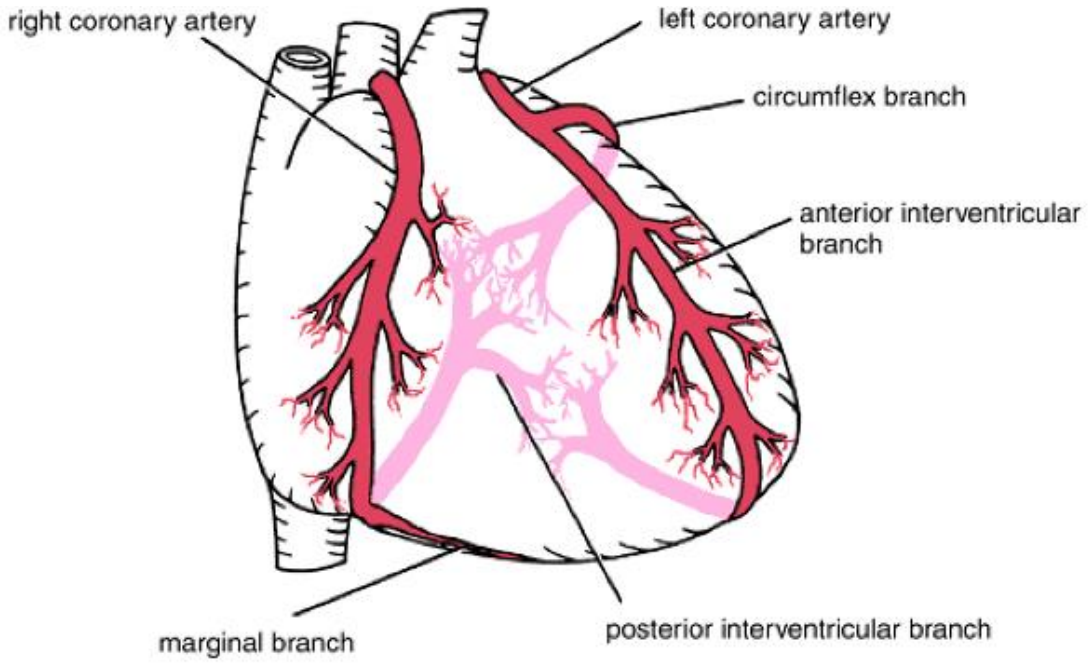


Fig. (12): Coronary arteries and their main branches .

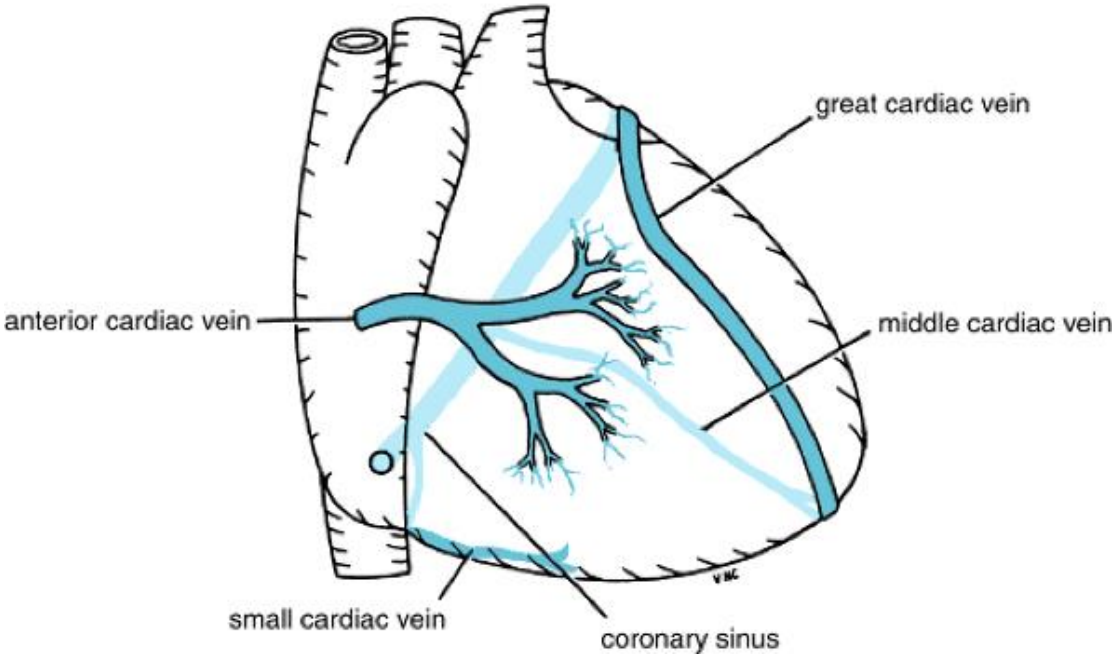


Fig. (13): Cardiac veins.

Note:

There are some veins open directly into the right atrium.

- *Anterior cardiac veins which they are three or four veins which run on the front of the right ventricle and open directly into the right atrium.*
- *Venae cordis Minimi: They are a number of minute veins from the wall of the heart opening directly into the right and left atria with fewer draining into the right and left ventricles.*

Lymphatic drainage of the heart:

The lymphatics of the heart drain into the tracheobronchial lymph nodes and the mediastinal lymph trunks.

The structure of the heart wall:

The wall of the heart consists of three layers (the epicardium, the myocardium and the endocardium).

The epicardium: It is the outermost layer of the heart. This is the visceral layer of the serous pericardium.

The myocardium: This is the intermediate layer of the heart wall. It is the main layer and consisting of cardiac muscle fibers. The myocardium varies in thickness between chambers, depending on the functional requirements of that chamber.

The endocardium: It is the inner lining of the cardiac chambers. It is a thin smooth endothelial layer.

The fibrous skeleton of the heart:

It consists of fibrous rings that surround the atrioventricular, pulmonary and aortic orifices. It provides attachment for the muscle fibers. The fibrous rings around the atrioventricular orifices separate the muscular wall of the atria from those of the ventricles. The function of these rings is to support the valvular opening and electrically isolates the ventricles from the atria.

Conducting system of the heart:

It consists of specialized cardiac muscle cells in certain regions of the myocardium have highly developed sensitivity and autorhythmicity. It includes the sinoatrial node, the atrioventricular node, the atrioventricular bundle and its right and left branches and the subendocardial plexus of purkinje fibers (Fig. 14).

The sinoatrial (S-A) node: It lies in the myocardium of the right atrium between the crista terminalis and the opening of the superior vena cava. It is called the pacemaker of the heart as it initiates the contraction cycle. Once initiated, the cardiac impulse spreads through the atrial myocardium to reach the atrioventricular node.

The atrioventricular (A-V) node: It is situated in the lower part of the atrial septum just above the attachment of the septal cusp of the tricuspid valve. It is not connected to the S. A. node by specialized conducting fibers, but it is stimulated by atrial depolarization. It gives rise to the atrioventricular bundle.

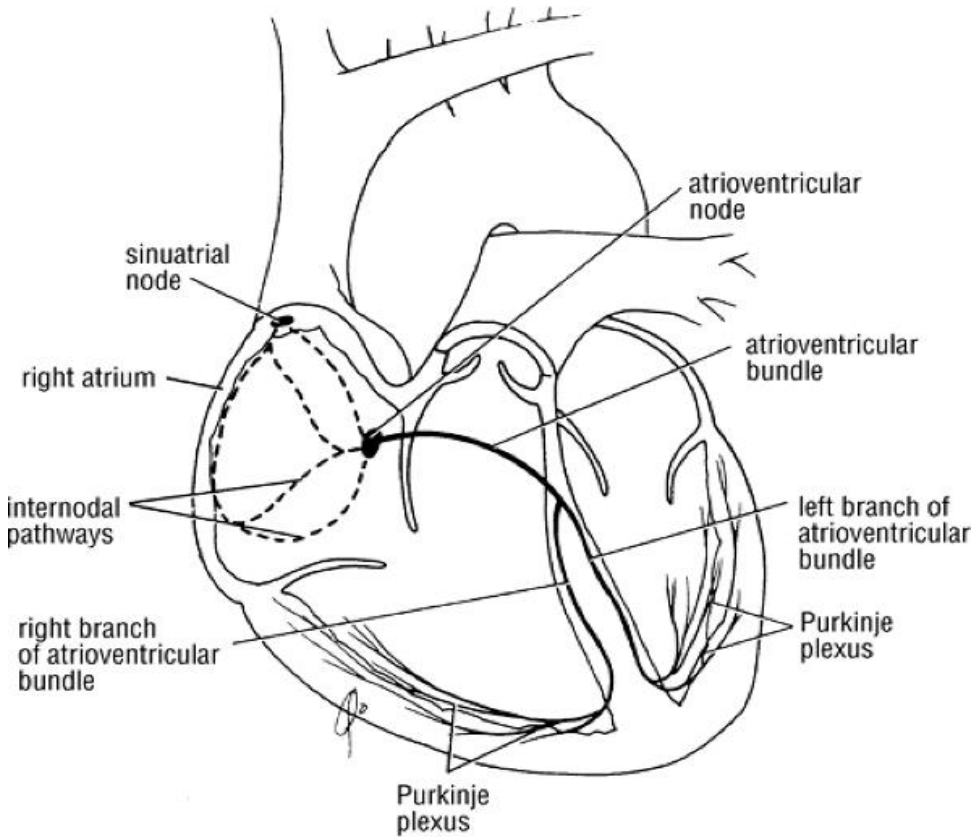


Fig.(14): Conducting system of the heart.

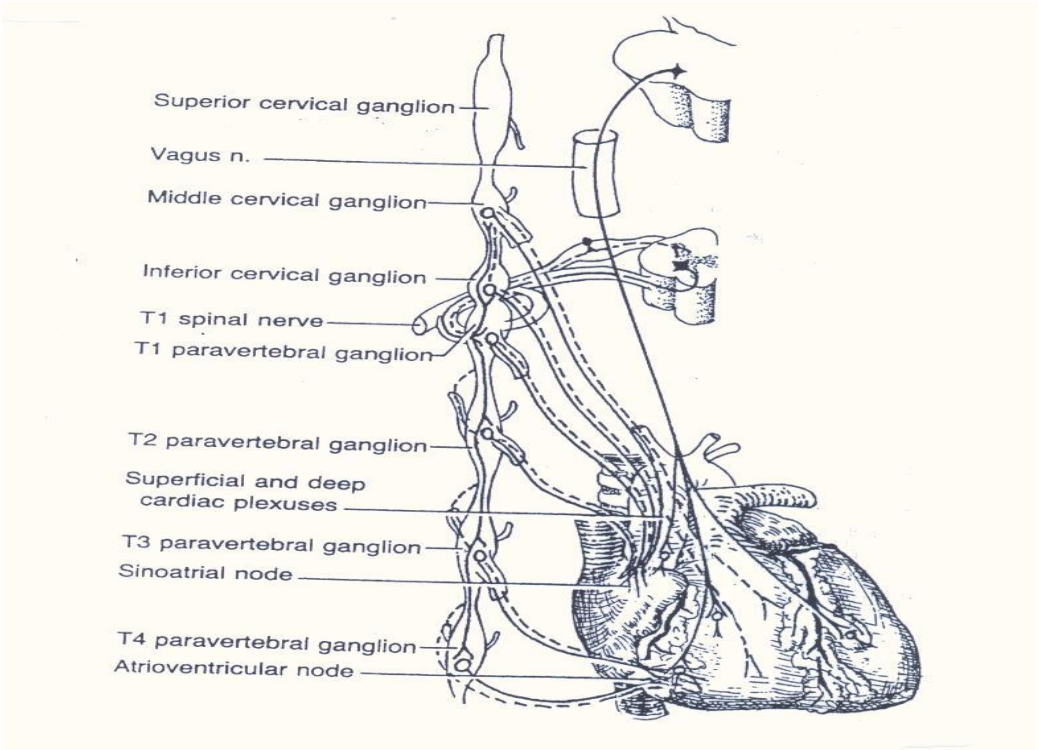


Fig. (15): Autonomic innervation of the heart .

The atrioventricular (A. V.) bundle:

It descends behind the septal cusp of the tricuspid valve to reach the inferior border of the membranous part of the ventricular septum. At the upper border of the muscular part of the septum it divides into right and left branches. The right branch passes down on the right side of the ventricular septum to reach the moderator band, then to the base of the anterior papillary muscle where it breaks into a plexus of purkinje fibers which are distributed to all parts of the right ventricle. The left branch runs in the septum beneath the endocardium of the ventricle towards the apex. It usually divides into two branches which give a plexus of purkinje fibers. This plexus is distributed beneath the endocardium to all parts of the left ventricle.

Nerve supply of the heart:

The heart is innervated by sympathetic and parasympathetic fibers of the autonomic nervous system via the cardiac plexuses. The sympathetic supply arises from the cervical and the upper thoracic portions of the sympathetic trunks, and the parasympathetic supply comes from the vagus nerves (Fig. 15).

Great vessels of the Thorax

A) Large veins of the thorax:

Brachiocephalic (innominate) veins:

These are two large veins **(right and left)** which drain the head and neck and upper limbs. Each vein is formed behind the corresponding sternoclavicular joint by the union of the internal jugular vein and the subclavian vein. **The right innominate vein** is about one inch long. It descends vertically close to the sternum to join the left innominate vein forming together the superior vena cava. **The left innominate vein** is longer than the right being about two and half inches long. It runs an oblique course downward and to the right to join the right innominate vein forming together the superior vena cava. It is related anteriorly to the manubrium sterni and the remains of the thymus gland. Posteriorly, it related to the large branches of the aorta (Fig. 16).

Superior vena cava

The superior vena cava is formed by the union of the two brachiocephalic veins. It passes downward close to the sternum to open into the right atrium of the heart. The vena azygos joins the superior vena cava just before it enters the pericardium.

Inferior vena cava

It enters the thorax by passing through the caval opening of the diaphragm opposite the eighth thoracic vertebra. It opens into the right atrium.

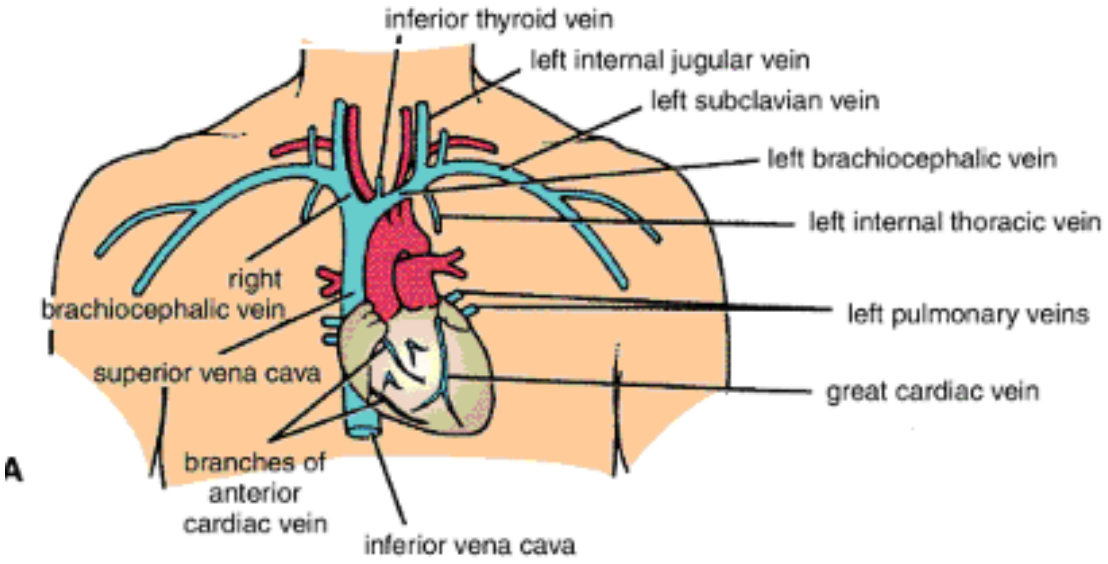


Fig. (16): It shows brachiocephalic , superior and inferior vena cava.

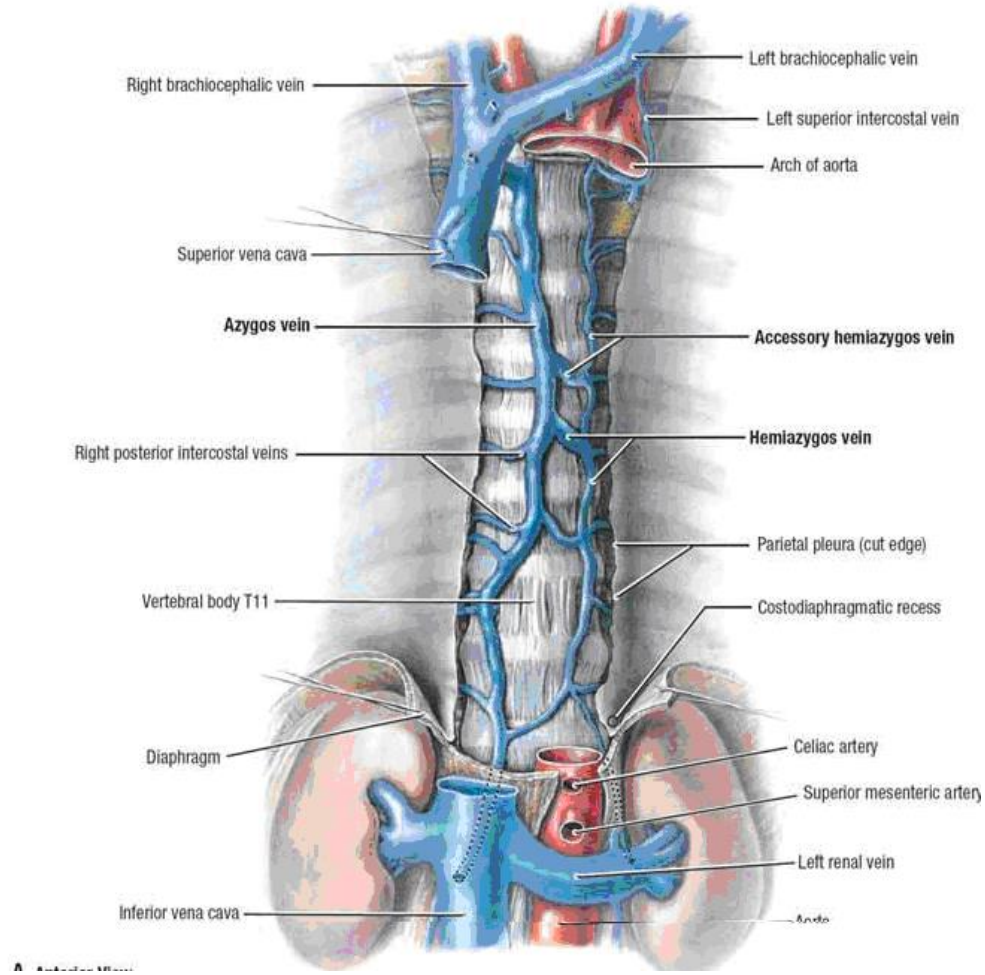


Fig. (17): Azygos system of veins .

The pulmonary veins:

There are two pulmonary veins leave each lung carrying oxygenated blood to the left atrium of the heart.

Azygos system of veins (Fig. 17)

- **Azygos vein**

It begins in the abdomen by the union of the right ascending lumbar vein with the right subcostal vein. The right ascending lumbar vein is joined by a small vein which arises from the back of the inferior vena cava and is known as the right lumbar azygos vein. So, the azygos vein acts as a large connection between the inferior and superior vena cava.

The azygos vein enters the thoracic cavity by passing through the aortic opening of the diaphragm. It ascends in the posterior mediastinum up to the level of the fourth thoracic vertebra where it arches above the root of the right lung to join the back of the superior vena cava.

The azygos vein receives the following tributaries:

1- The right posterior intercostal veins from the eleventh up to the fourth.

2- The right superior intercostal vein.

3- The hemiazygos vein.

4- The accessory hemiazygos vein.

5- The oesophageal veins.

6- The pericardial veins.

7- The mediastinal veins.

- **The hemiazygos (inferior) hemiazygos vein**

It begins in the abdomen by the union of the left ascending lumbar vein with the left subcostal vein. The left ascending lumbar vein is joined by the left lumbar azygos vein which arise from the back of the left renal vein

It enters the thorax by piercing the left crus of the diaphragm. It ascends on the left side of the bodies of the thoracic vertebra. Then, it crosses the body of the eighth thoracic vertebra to join the vena azygos. It receives the following tributaries:

- The left posterior intercostals veins from number nine to eleven.
- Oesphageal veins.
- Mediastinal veins.

- **The superior (accessory) hemiazygos vein**

It is a longitudinal channel descends in the posterior mediastinum. It crosses the body of the eighth thoracic vertebra to join the vena azygos. It receives the following tributaries:

- The left posterior intercostals vein from number four to eight.
- Bronchial veins from the left lung.
- Oesophageal veins.

Large arteries of the Thorax:

Aorta:

This is the main arterial trunk which carries oxygenated blood to all parts of the body. Its course is divided into:

- Ascending aorta.
- Arch of aorta.
- Descending aorta which is subdivided into descending thoracic aorta and abdominal aorta.

• Ascending aorta:

It begins from the vestibule of the left ventricle and runs upward and forward to become behind the right half of the sternal angle where it becomes continuous with the arch of the aorta. It is about two inches long. The ascending aorta and the pulmonary trunk is enclosed in a single sheath of serous pericardium. It gives the following branches:

1- The right coronary artery which arises from the anterior aortic sinus.
2- The left coronary artery which arises from the left posterior aortic sinus.

• Arch of aorta:

The arch of the aorta is a continuation of the ascending aorta. It begins behind the right border of the sternum at the level of the upper border of the right second costal cartilage.

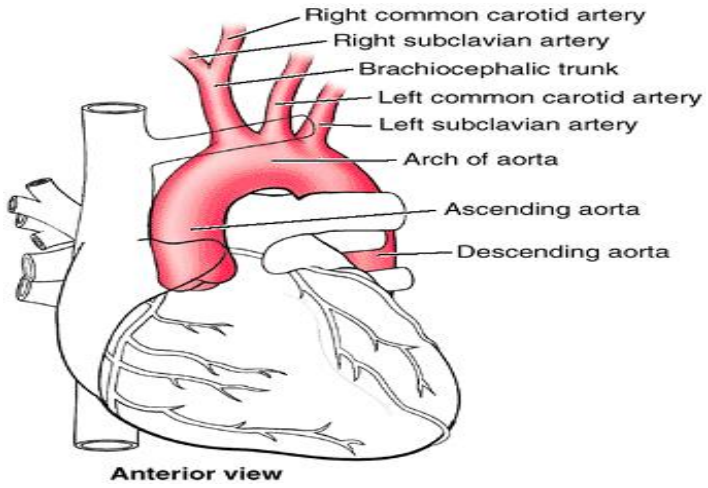


Fig.(18): It shows branches of the aortic arch

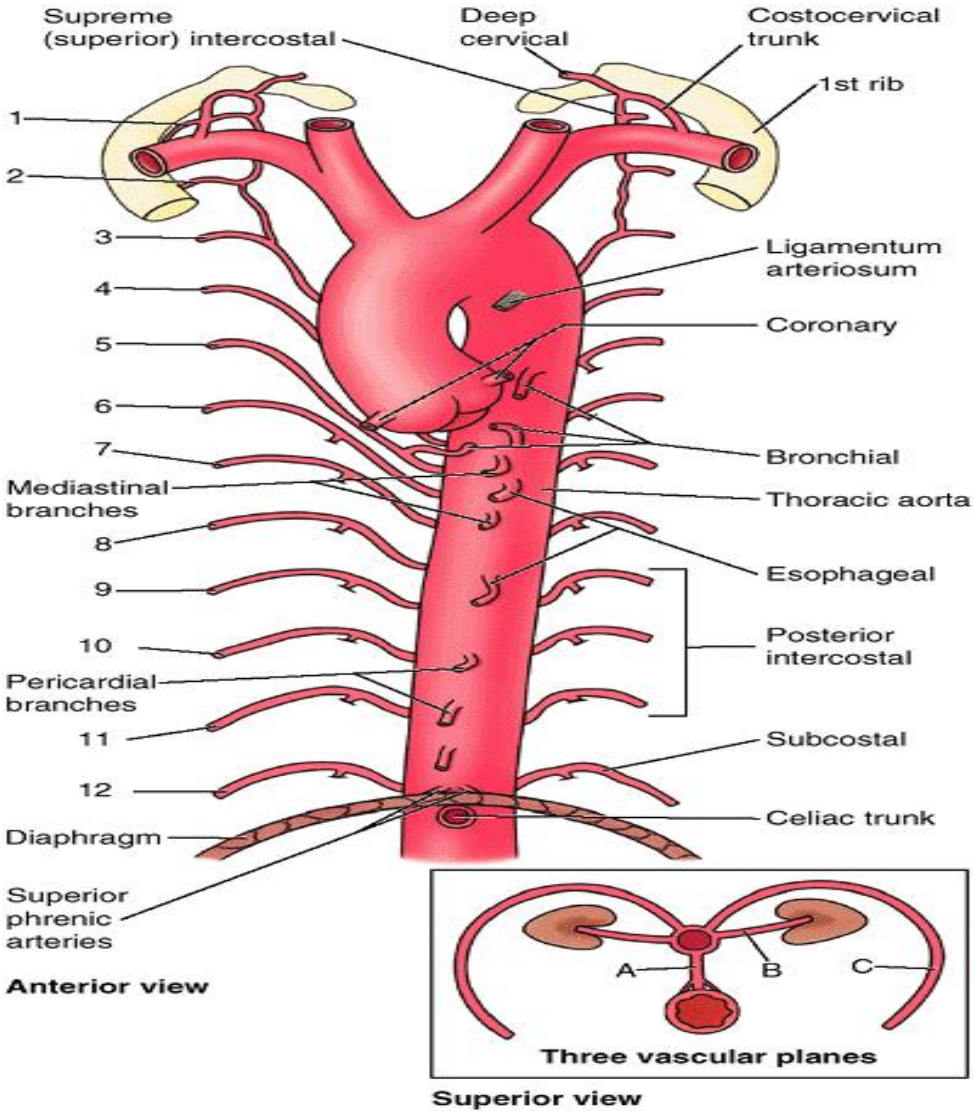


Fig. (19): Main branches of the ascending, arch and descending aorta.

It runs an arched course behind the lower half of the manubrium sterni. At first, it passes upwards, backwards and to the left in front of the trachea, then it curves backwards on the left side of the trachea. After this, it curves downwards to reach the left side of the lower border of the body of the fourth thoracic vertebra where it continues as the descending thoracic aorta. It gives the following branches (Fig.18):

1-**The brachiocephalic (Innominate) artery**: It arises from the arch of the aorta opposite the center of the manubrium sterni. Then it passes upward, backward and to the right and it ends at the right sternoclavicular joint where it divides into the right common carotid and the right subclavian arteries.

2-**The left common carotid artery**: It arises from the aortic arch close to the left side of the innominate artery. It ascends upwards, backwards and to the left, passing behind the left sternoclavicular joint where it continues up into the neck.

The left subclavian artery: It arises from the aortic arch behind the left common carotid artery. It runs upward along the left side of the trachea and the oesophagus to enter the root of the neck.

Descending thoracic aorta

It begins as a continuation of the arch of the aorta on the left side of the lower border of the body of the fourth thoracic vertebra (opposite the sternal angle). It extends downward in the posterior mediastinum to the level of the twelfth thoracic vertebra where it passes through the aortic opening of the diaphragm. It continues into the abdomen as the abdominal aorta. It gives the following branches (Fig. 19).

- Posterior intercostal arteries: It passes to the lower nine intercostal spaces on each side.
- Subcostal arteries: It runs along the lower border of the twelfth rib on each side.
- Pericardial branches.
- Oesophageal branches.
- Left bronchial arteries.
- Mediastinal branches.

The pulmonary trunk:

The pulmonary trunk conveys deoxygenated blood from the right ventricle of the heart to the lungs. It arises from the upper part of the infundibulum of the right ventricle and runs upward, backward and to the left. It is about 2 inches long and terminates in the concavity of the aortic arch by dividing into right and left pulmonary arteries.

Lymphatic drainage of the thorax

Thoracic wall:

- The lymphatics of the skin of the anterior thoracic wall drain to the anterior axillary lymph nodes. The lymphatics of the skin of the posterior thoracic wall drain to the posterior axillary nodes.
- The deep lymphatics of the anterior parts of the intercostals spaces drain forward to the internal thoracic nodes along the internal thoracic blood vessels. From these lymph nodes, lymphatic vessels drain into the thoracic duct on the left side

and into the bronchomediastinal lymph trunk on the right side.

- The deep lymphatics of the posterior parts of the intercostals spaces drain backward to the posterior intercostal nodes lying near the heads of the ribs. From these nodes, lymph vessels drain into the thoracic duct.

Lungs.

Lymph from the lungs passes through the superficial and deep lymphatic plexus of vessels to the bronchopulmonary lymph nodes which drain into the bronchomediastinal lymph trunk.

Heart

The lymphatics of the heart drain along its blood vessels into the tracheobronchial lymph nodes and mediastinal lymph trunks.

Mediastinum.

There are several lymph nodes scattered through the mediastium. They drain lymph from mediastinal structures and empty into the bronchomediastinal trunks and the thoracic duct.

Thoracic duct

This duct drains all the lymph of the body with the exception of the right side of the head and neck, right upper limb and the right side of the thorax (Fig. 20).

- The thoracic duct begins in the abdomen from the upper end of the cisterna chyli. It enters the thorax by passing through the aortic opening in the diaphragm.
- It continues through the posterior mediastinum and gradually crosses the median plane behind the oesphagus to the left side at the level of the fifth thoracic vertebra. It then runs upward along the left edge of the oesophagus to enter the root of the neck.
- In the root of the neck, it arches laterally behind the carotid sheath and infront of the vertebral vessels. Then, it curves downwards in the front of the first part of the subclavian artery and ends in the angle between the left internal jugular and the left subclavian veins where they join to form the left innominte vein.
- The duct contains several valves. It receives the following tributaries:

1-Lymphatics drain the left thoracic wall and a small portion of the left breast.

2-At the root of the neck, it receives the left jugular lymph trunk which carries lymph from the left side of the head and neck.

3-It receives the left subclavian lymph trunk which carries lymph from the left upper limb.

4-It also receives the left bronchomediastinal lymph trunk .

Notes

At the root of the neck ,the left jugular ,the left subclavian and the left bronchmedistinal lymph trunks usually drain into the thoracic duct .Sometimes they drain directly into the adjacent large veins .

The right jugular lymph trunk which drain the right side of the head and neck , the right subclavian lymph trunk which drain the right upper limb and the right bronchomediastinal lymph trunk which drain the right side of the thorax usually open into the adjacent great veins. Sometimes, the right jugular, right subclavian and the right bronchomediastinal lymph trunks may join to form the right lymphatic duct. This common duct, if present is about ½ inch long and opens into the beginning of the right brachiocephalic vein.

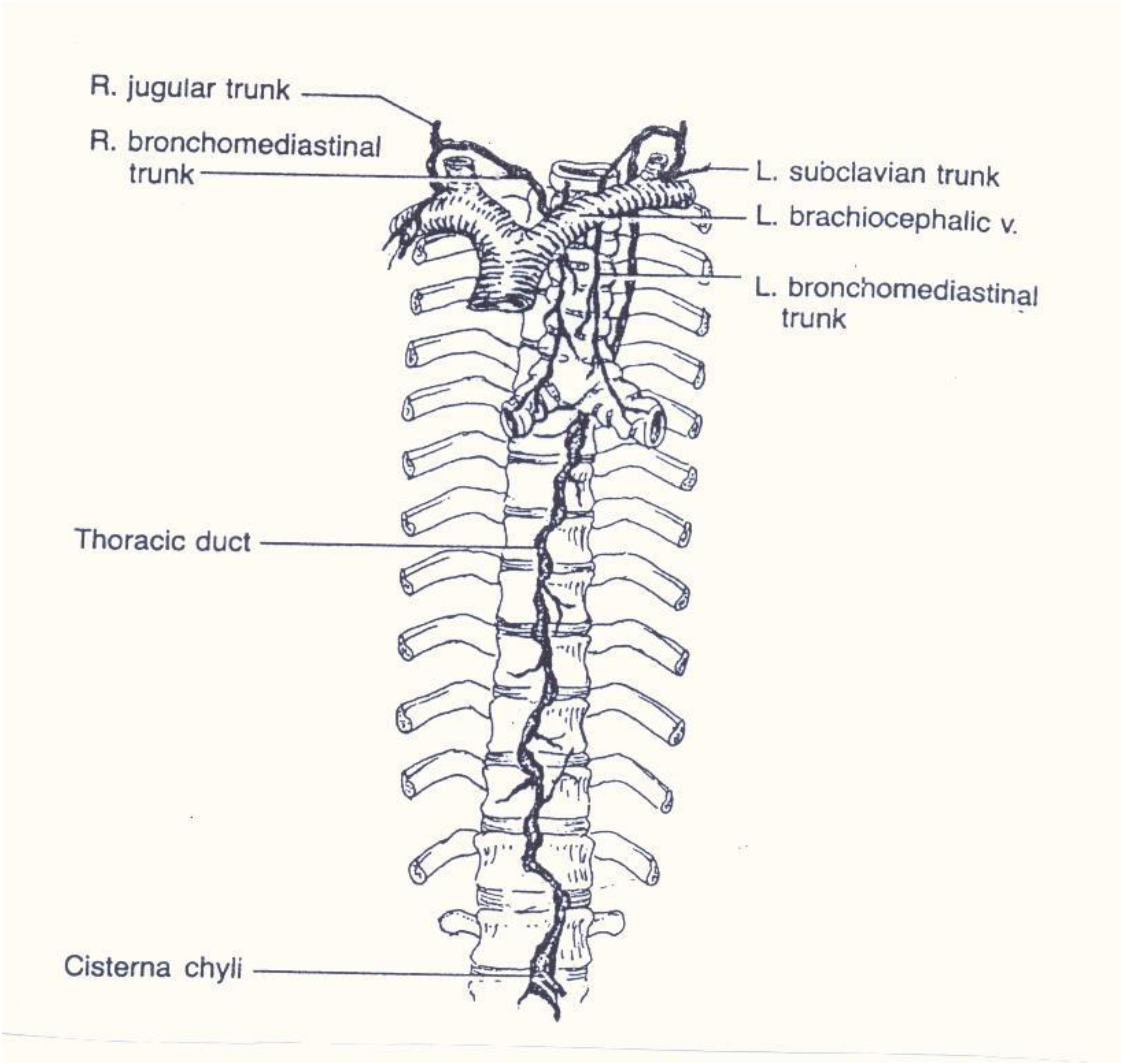


Fig. (20): Thoracic duct .

Nerves of the Thorax

Phrenic nerves:

The phrenic nerves arise in the neck from the anterior rami of the third, fourth and fifth cervical nerves.

The right phrenic nerve: It descends in the thorax along the right side of the right brachiocephalic vein and the superior vena cava. It passes in front of the root of the right lung and runs along the right side of the pericardium. Then, it descends on the right side of the inferior vena cava to reach to the diaphragm.

The left phrenic nerve: It descends in the thorax along the left side of the left subclavian artery. It crosses the left side of the aortic arch and passes in front of the root of the left lung. Then it descends over the left surface of the pericardium to reach the diaphragm

Note:

The right and left phrenic nerves are the main motor nerve supply to the diaphragm.

Autonomic nerves of the thorax (Fig. 21)

I) Thoracic part of the sympathetic trunk:

The thoracic part of the sympathetic trunk is continuous above with the cervical and below with the lumbar parts of the sympathetic trunk. It has twelve segmentally arranged ganglia. Each ganglia has a white and gray ramus communicans passing to the corresponding spinal nerve. The first ganglion is often fused with the inferior cervical ganglion to form the stellate ganglion.

It runs downward on the heads of the ribs. It leaves the thorax on the side of the body of the twelfth thoracic vertebra by passing behind the medial arcuate ligament. It gives innervation to the following structures:

1-It gives gray rami communicantes to all the thoracic spinal nerves. The postganglionic fibers are distributed through the branches of the spinal nerves to the blood vessels, sweat glands and arrector pili muscles of the skin.

2-the first five ganglia give postganglionic fibers that share in the cardiac, pulmonary and oesophageal plexuses.

3-The lower eight ganglia mainly give preganglionic fibers which are grouped together to form the splanchnic nerves. The greater splanchnic nerve arises from ganglia 5-9, the lesser splanchnic nerve arises from ganglia 10 and 11, and the lowest splanchnic nerve arises from ganglion 12. They enter the abdomen by piercing the crura of the diaphragm to supply the abdominal viscera.

II)Vagus nerves

The right vagus nerve:

It descends into the thorax lying posterolateral to the brachiocephalic artery, then lateral to the trachea and medial to the terminal part of the azygos vein. It passes behind the root of the right lung and assists in the formation of the pulmonary plexus. The vagus passes onto the posterior surface of the oesphagus and share in the formation of the esophageal plexus. It then passes through the esophageal opening in the diaphragm to reach the posterior surface of the stomach forming the posterior gastric nerve.

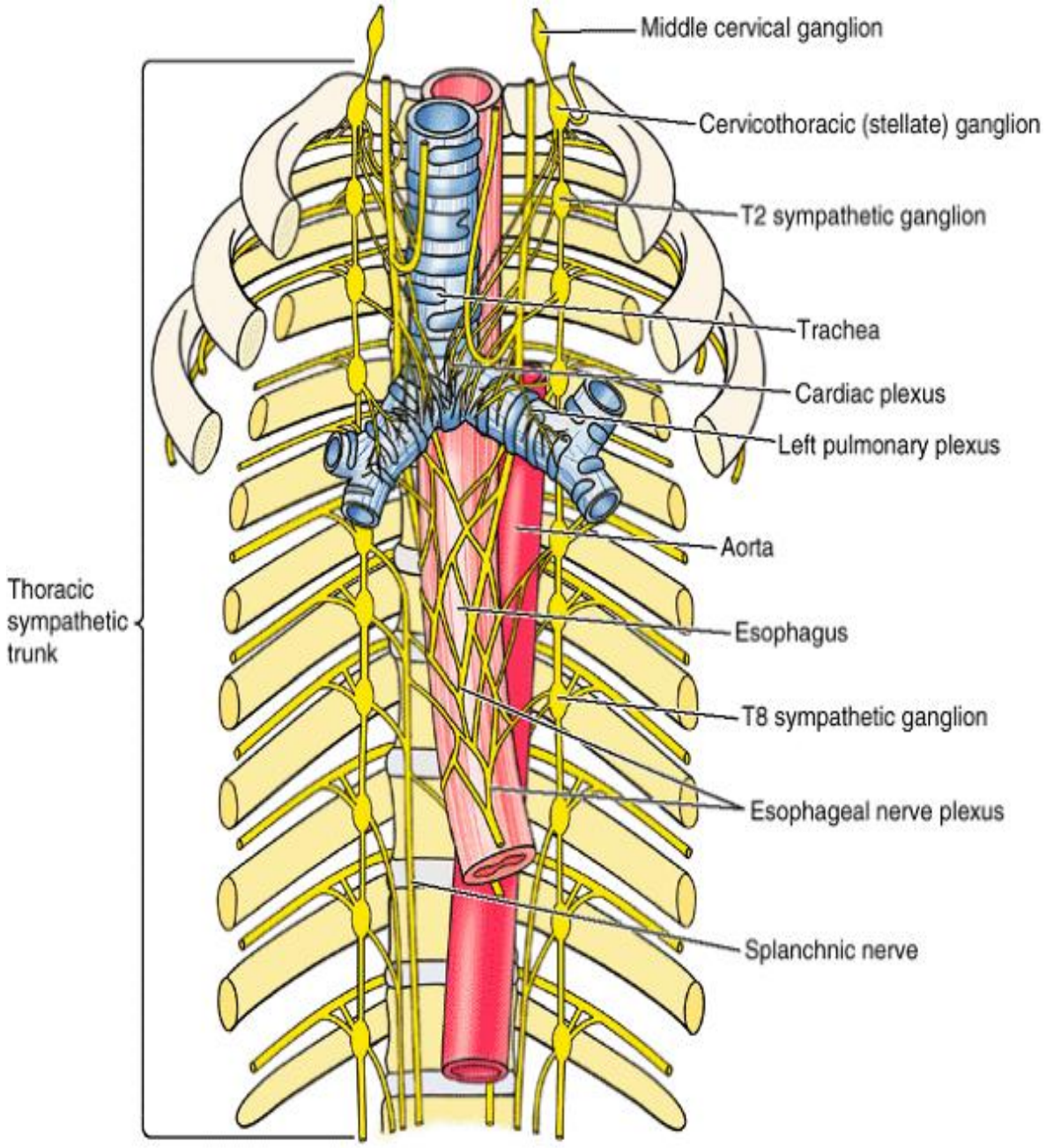


Fig.(21): Autonomic nerves of the thorax .

The left vagus nerve:

It descends in the thorax between the left common carotid and left subclavian arteries. Then, it crosses the left side of the aortic arch and turns backwards behind the root of the left lung and assists in the formation of the pulmonary plexus. After this, it passes onto the anterior surface of the esophagus and shares the formation of esophageal plexus. Then, it passes through the esophageal opening in the diaphragm to reach the anterior surface of the stomach forming the anterior gastric nerve.

Branches:

- The right and left vagus nerves give branches that share in the cardiac, pulmonary and esophageal plexuses of nerves.
- The left recurrent laryngeal nerve arises from the left vagus nerve on the left side of the aortic arch. It curves below the aortic arch and behind the ligamentum arteriosum. Then, it ascends upwards in the groove between the trachea and esophagus to reach the larynx.

Trachea

The trachea is a tube about 5 inches (13 cm) long and one inch (2.5cm) in diameter.

- It begins in the neck at the level of the sixth cervical vertebra (opposite the cricoid cartilage)and ends at the level of the lower border of the fourth thoracic vertebra (opposite the sternal angle) by dividing into the right and left principal bronchi (Fig. 22). In deep inspiration ,the bifurcation descends to the level of the sixth thoracic vertebra.
- The trachea has a fibroelastic wall which is supported by cartilaginous rings which are incomplete posteriorly.

Relations of the trachea:

The trachea has a cervical and thoracic course. In the thorax, it is related anteriorly to the sternum, the thymus, the left brachiocephalic vein, the origins of the brachiocephalic, left common carotid arteries and the arch of the aorta.

Posteriorly, the trachea is related to the esophagus and the left recurrent laryngeal nerve. On the right side, it is related to the azygos vein, the right vagus nerve and the right pleura and lung. On the left side, it is related to the arch of the aorta, left common carotid, left subclavian artery and the left pleura and lung.

Arterial supply of the trachea: It is mainly derived from the inferior thyroid arteries and its lower end is supplied by the bronchial arteries.

Nerve supply of the trachea: It receives autonomic nerve fibers from the vagi and the sympathetic trunks.

Principal bronchi:

- **The right principal bronchus** is wider, shorter and more vertical than the left. It is about 1 inch (2.5 cm) long. It gives the superior lober bronchus before entering the hilum of the right lung. On entering the hilum, it divides into a middle and inferior lober bronchus.
- **The left principal bronchus** is narrower longer and more horizontal than the right. It is about 2 inches (5 cm) long. It passes to the left below the arch of the aorta and in front of the esophagus. On entering the hilum of the left lung, it divides into a superior and an inferior lobar bronchus.

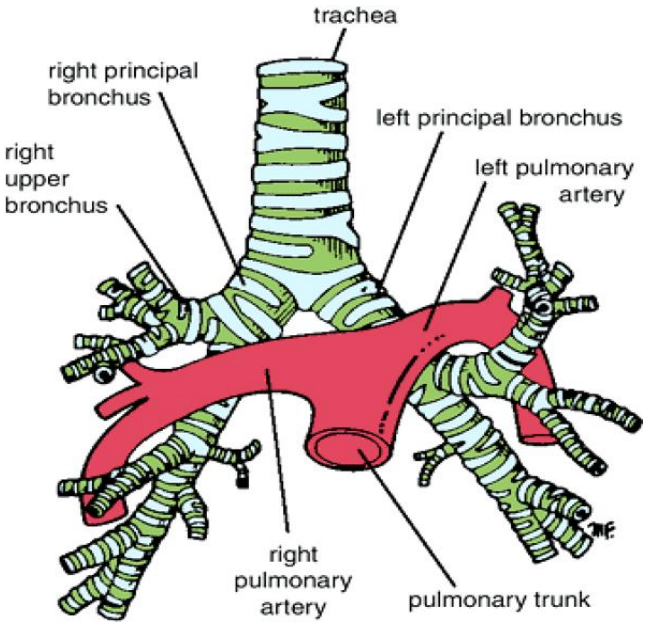


Fig. (22): Trachea and its main bronchi .

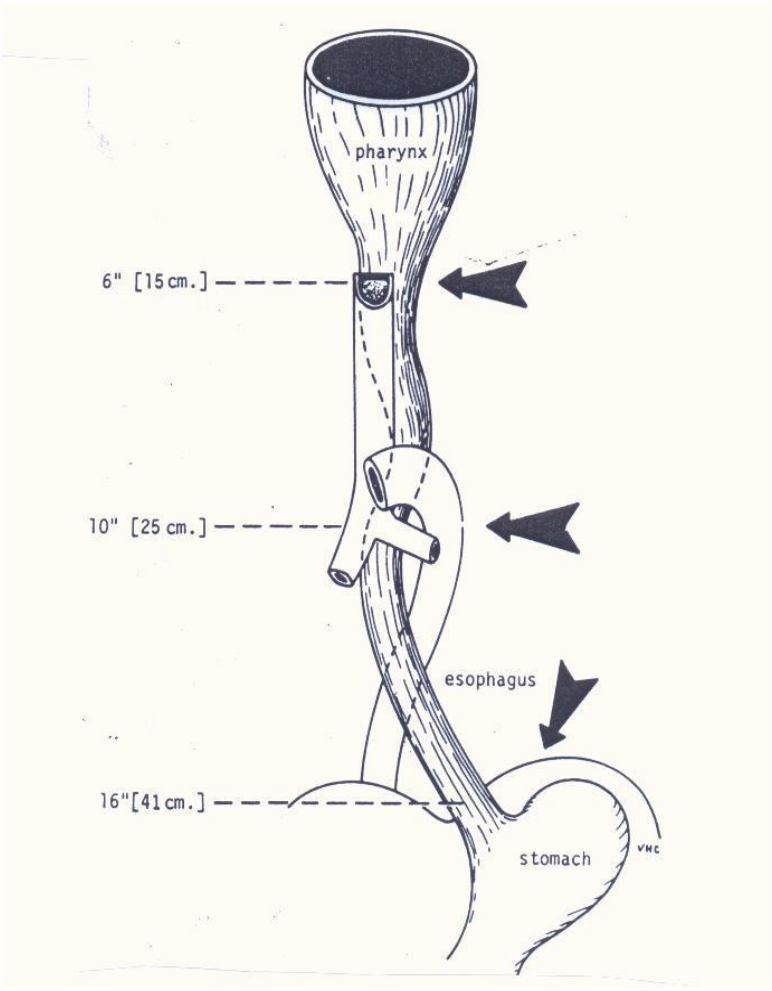


Fig. (23): It shows the oesophagus and the sites of its constrictions

Oesophagus

The oesophagus is a muscular tube about 10 inches (25 cm) long. It extends from the lower end of the pharynex to the cardiac end of the stomach. It begins in the neck at the level of the sixth cervical vertebra (opposite the lower border of the cricoid cartilage). Then, it descends into the thorax and pass through the diaphragm (at the level of the tenth thoracic vertebra) to join the stomach (Fig. 23).

The oesophagus passes from the neck downward into the thorax through the superior then the posterior mediastina. The thoracic part of the oesophagus has the following relations.

Anteriorly, the oesophagus is related to the trachea and the left recurrent laryngeal nerve, the left principal bronchus and the pericardium which separates it from the left atrium.

Posteriorly. It is related to the bodies of the thoracic vertebrae, the thoracic duct, the azygos veins and the right posterior intercostal arteries. Its lower end is related to the descending thoracic aorta.

On the right side: It is related to the right pleura and lung and to the terminal part of the azygos vein.

On the left side: It is related to the left subclavian artery, the aortic arch, the thoracic duct and the left pleura and lung.

Notes:

The oesophagus has three anatomical contrictions. ***The first*** *is where the pharynx joins its upper ends,* ***the second*** *is where the aortic arch and the left bronchus cross its anterior surface and* ***the third***

presents where the oesphagus passes through the diaphragm into the stomach.

The oesophagus at its beginning lies in the median plane. As it descends in the neck, it deviates slightly to the left. In the superior mediastinum it inclines again towards the median plane which it reaches opposite the fifth thoracic vertebra. Opposite the seventh thoracic vertebra it deviates to the left once more and pierces the diaphragm one inch to the left of the median plane at the level of the tenth thoracic vertebra.

Blood supply of the oesophagus:

A) Arterial supply:

- The upper third of the oesophagus is supplied by branches from the inferior thyroid artery.
- The middle third is supplied by branches from the descending thoracic aorta.
- The lower third is supplied by branches from the left gastric artery.

B) Venous drainage.

- The veins from the upper third of the oesophagus drain into the inferior thyroid vein.
- The veins from its middle third drain into the azygos vein.
- The veins from its lower third drain into the left gastric vein which it is a tributary of the portal vein.

Nerve supply:

The oesophagus is supplied by autonomic nerve fibers from the vagi and sympathetic trunk through the oesophageal nerve plexus.